

RYOTARO NIWA

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RESEARCH INTEREST

Quantum Information Theory

Quantum process learning, quantum algorithms, quantum error-correction

EDUCATION

PhD

April 2026-Present

University of Tokyo, Supervisor: Prof.Mio Murao

MSc

April 2024-March 2026

University of Tokyo, Supervisor: Prof.Mio Murao

Thesis: “Quantum algorithms for learning the spectrum of quantum channels”

BSc

April 2020-March 2024

University of Tokyo, Major: Physics

PUBLICATION & PREPRINTS

* denotes first author

[1] **R.Niwa***, S.Yoshida, K.Ono, T.Utsumi, Z.Li, Y.Yang, R.Takagi and M.Murao, *Scaling-optimal purification of noisy qubit unitary channels*, [arXiv:2606.12394](#) (2026)

[2] S.Yoshida, **R.Niwa**, and M.Murao, *Random dilation superchannel*, [arXiv:2512.21260](#) (2025)

[3] **R.Niwa***, Z.M.Rossi, P.Taranto, and M.Murao, *Singular value transformation for unknown quantum channels*, [arXiv:2506.24112](#) (2025)

[4] **R.Niwa*** and J.Y.Lee, *Coherent information for Calderbank-Shor-Steane codes under decoherence*, [PhysRevA.111.032402](#) (2025)

RESEARCH TALK & SEMINARS

- “Scaling-optimal purification of noisy qubit unitary channels”, Seminar talk at the Technical University of Munich, June 2026
- “Singular value transformation for unknown quantum channels”, Seminar talk at the University of Edinburgh, February 2026

PRESENTATION

Contributed Posters

- 7th Seefeld Workshop on Quantum Information, Seefeld, June 2026
- 29th Annual Quantum Information Processing Conference 2026 (QIP 2026), Riga, January 2026
- 25th Asian Quantum Information Science Conference (AQIS 2025), Hong Kong, August 2025
- Quantum Innovation 2025, Osaka, July 2025
- From quantum materials to quantum information (QMQUI 2024), Okinawa, November 2024
- Quantum Innovation 2024, Tokyo, October 2024

AWARDS AND FELLOWSHIPS

Excellent Research Assistant
Q-leap Quantum Information Processing

2024-2026

Scholarship Award
Scholarship for the Study and Visit Abroad Program (SVAP)

September 2023

TEACHING EXPERIENCES

Quantum Mechanics II
Teaching Assistant (TA)

Spring 2024

Exercise in physics I
Teaching Assistant (TA)

Fall 2024

COMMUNITY SERVICE

Reviewer for Quantum